

Design and Control of a Three-Phase Grid-Connected Inverter

Simulation and Control Implementation

Yashwardhan and Shahnawaz Shahid

M.Tech, Electrical Engineering IIT

Kanpur

Project Report

Abstract

This project presents the modelling, control, and simulation of a three-phase grid-connected inverter. The main objective of the system is to control the exchange of power between a DC source and the electrical grid while maintaining synchronization with the grid voltage.

The developed system consists of a DC-link source, a three-phase voltage source inverter, an interface filter, and a three-phase grid. A synchronous reference frame control strategy is implemented using Clarke and Park transformations. A phaselocked loop (PLL) is used to estimate the grid angle required for synchronization and coordinate transformation.

The grid currents are controlled in the dq reference frame using PI controllers. The d -axis current is associated primarily with active power control, whereas the q -axis current is associated primarily with reactive power control. The controller generates reference voltage signals, which are transformed back into the three-phase reference frame and used to generate PWM gate pulses for the inverter switches.

Simulation results demonstrate the synchronization of the inverter with the grid, generation of three-phase inverter voltages, controlled grid current injection, and the operation of the complete control system.

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1 Introduction

Grid-connected power electronic converters are widely used in renewable energy systems, energy storage systems, electric vehicle charging systems, and distributed generation. A grid-connected inverter converts DC electrical power into AC power and injects the controlled power into the utility grid.

Unlike a standalone inverter, a grid-connected inverter must operate in synchronization with the grid. Therefore, the inverter control system must accurately determine the phase angle and frequency of the grid voltage. In addition, the injected current must be controlled to achieve the desired active and reactive power exchange.

In this project, a three-phase grid-connected inverter model is developed and simulated. The complete system includes the power circuit, grid synchronization using a phase-locked loop, coordinate transformations, PI-based current controllers, modulation signal generation, and PWM gate pulse generation.

2 Objectives

The main objectives of this project are:

- To model a three-phase voltage source inverter.
- To connect the inverter to a three-phase AC grid through an interface filter.
- To synchronize the inverter control system with the grid voltage.
- To implement abc to dq coordinate transformations.
- To control the d -axis and q -axis components of the grid current.
- To implement PI controllers for current regulation.
- To generate three-phase modulation signals.
- To generate PWM gate pulses for the inverter switches.
- To analyze inverter voltage, grid voltage, grid current, and dq -axis quantities.

3 System Configuration

The overall system consists of the following major components:

1. DC source or DC-link.
2. Three-phase voltage source inverter.
3. Grid interface filter.
4. Three-phase AC grid.
5. Voltage and current measurement blocks.
6. Phase-locked loop for grid synchronization.
7. abc - dq and dq - abc coordinate transformation blocks.

8. PI-based current controllers.
9. PWM generation and gate pulse generation.

The basic power flow of the system is illustrated as:

DC Source → Inverter → Filter → Grid

The controller measures the grid voltages and currents and generates appropriate switching signals for the inverter.

4 Power Circuit of the Grid-Connected Inverter

The power circuit consists of a DC-link source connected to a three-phase inverter bridge. The inverter generates a controlled three-phase AC voltage by appropriately switching the power semiconductor devices.

The inverter output is connected to the utility grid through an interface filter. The purpose of the filter is to reduce the switching harmonics produced by the PWM operation of the inverter.

Figure 1 shows the complete power circuit used in the simulation.

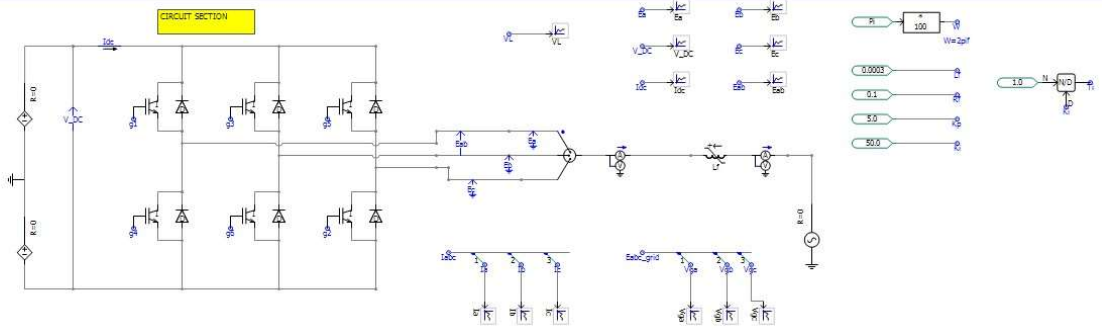


Figure 1: Circuit section of the three-phase grid-connected inverter.

The main components visible in the circuit are the DC source, three-phase inverter bridge, interface inductance, measurement blocks, and the three-phase grid connection.

5 Operating Principle of the Three-Phase Inverter

A three-phase voltage source inverter consists of three switching legs. Each leg generates one phase voltage. The switching states of the upper and lower switches are complementary to prevent a short circuit across the DC link.

The switching operation generates the inverter phase voltages:

$$V_{ia}, \quad V_{ib}, \quad V_{ic} \quad (1)$$

The inverter output contains a fundamental AC component along with high-frequency switching harmonics. Therefore, an interface inductor or filter is used between the inverter and the grid.

The current through the interface inductor can be represented approximately as:

$$L \frac{di}{dt} = v_{\text{inv}} - v_{\text{grid}} - Ri \quad (2)$$

where:

- L is the interface inductance,
- R represents the interface resistance,
- v_{inv} is the inverter voltage,
- v_{grid} is the grid voltage,
- i is the grid current.

By controlling the inverter voltage, the current injected into the grid can be controlled.

6 Control Structure

The control system is responsible for generating the switching signals required to regulate the inverter current and synchronize the inverter with the grid. The major control stages are:

1. Measurement of three-phase grid voltages and currents.
2. Grid synchronization using PLL.
3. Transformation of measured quantities from the abc frame to the synchronous dq frame.
4. Comparison of measured currents with reference currents.
5. PI current control.
6. Generation of reference voltage signals.
7. Transformation of reference voltages from dq to abc .
8. Generation of modulation signals.
9. PWM generation.
10. Generation of gate pulses for the six inverter switches.

The complete control implementation is shown in Figure 2.

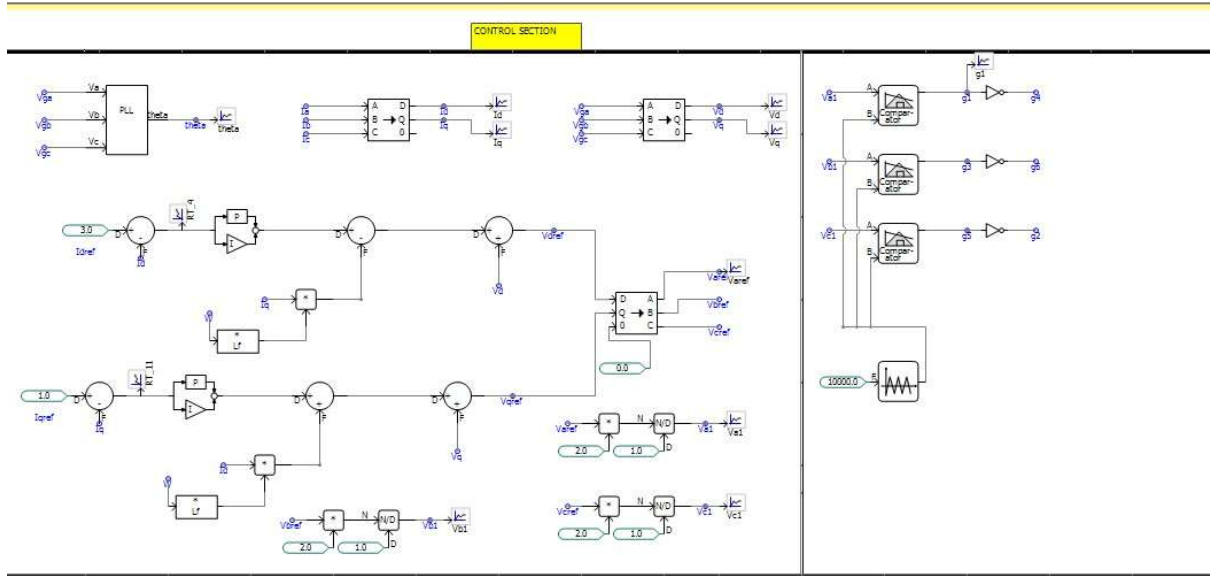


Figure 2: Control section implemented for the grid-connected inverter.

7 Grid Synchronization Using PLL

For a grid-connected inverter, synchronization with the grid is essential. The controller must determine the instantaneous phase angle of the grid voltage. A phase-locked loop is used to estimate the grid angle:

$$\theta \quad (3)$$

The estimated angle is then used for the coordinate transformations.

In a synchronous reference frame PLL, the three-phase grid voltage is transformed into the rotating dq reference frame. The PLL attempts to align the d -axis with the grid voltage vector. Under ideal synchronization:

$$v_q \approx 0 \quad (4)$$

and the d -axis voltage represents the grid voltage magnitude.

The estimated angle can be expressed as:

$$\theta = \int \omega dt \quad (5)$$

where ω is the estimated angular frequency.

8 Coordinate Transformation

The three-phase quantities are transformed into a rotating reference frame to simplify the control problem.

8.1 Clarke Transformation

The three-phase quantities are first transformed from the abc frame to the stationary $\alpha\beta$ frame.

$$\begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} \quad (6)$$

8.2 Park Transformation

The stationary frame quantities are transformed into the rotating dq frame using the estimated grid angle θ .

$$\begin{bmatrix} x_d \\ x_q \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} \quad (7)$$

The main advantage of the dq transformation is that balanced sinusoidal quantities appear approximately as DC quantities in the synchronous reference frame.

9 d -Axis and q -Axis Current Control

The measured three-phase grid currents are transformed into the synchronous dq reference frame.

The current components are:

$$i_d \quad (8)$$

and

$$i_q \quad (9)$$

Reference currents are defined as:

$$i_d^* \quad (10)$$

and

$$i_q^* \quad (11)$$

The current errors are:

$$e_d = i_d^* - i_d \quad (12)$$

$$e_q = i_q^* - i_q \quad (13) \text{ These errors are processed by PI controllers.}$$

10 PI Current Controller

The PI controller is used to eliminate the steady-state error between the reference current and the measured current.

The general PI controller is:

$$G_{PI}(s) = K_p + \frac{K_i}{s} \quad (14)$$

Therefore, the controller output can be written as:

$$u(t) = K_p e(t) + K_i \int e(t) dt \quad (15)$$

Separate PI controllers are used for the d -axis and q -axis current control loops. The controller generates voltage reference signals:

$$v_d^* \quad (16)$$

and

$$v_q^* \quad (17)$$

These reference voltages are subsequently transformed back into the three-phase reference frame.

11 Inverse dq Transformation

The voltage references generated by the current controllers are transformed from the rotating dq reference frame back into the stationary reference frame. The inverse Park transformation is:

$$\begin{bmatrix} v_\alpha^* \\ v_\beta^* \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} v_d^* \\ v_q^* \end{bmatrix} \quad (18)$$

The resulting reference voltages are then converted into three-phase reference signals:

$$v_a^*, \quad v_b^*, \quad v_c^* \quad (19)$$

12 PWM Generation

The three-phase reference voltage signals are compared with a high-frequency carrier waveform to generate PWM switching signals.

The modulation signals determine the duty cycle of the inverter switches.

The PWM generator produces six gate signals corresponding to the six power switches of the three-phase inverter.

The upper and lower switches in each inverter leg operate in a complementary manner.

13 Simulation Results

This section presents the major simulation results obtained from the grid-connected inverter model.

13.1 DC-Link Voltage and Current Components

Figure 3 shows the simulation results of the current and voltage components in the synchronous reference frame.

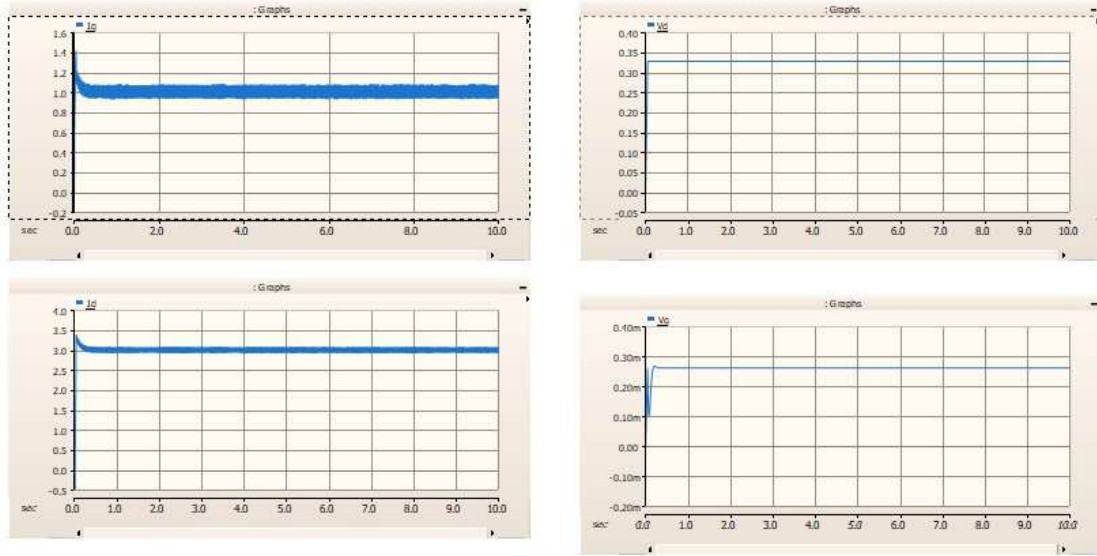


Figure 3: DC-link and dq -frame voltage/current quantities.

The figure shows the behavior of the measured quantities after the initial transient period. The dq transformation converts the three-phase sinusoidal quantities into controlled components in the synchronous reference frame.

13.2 Gate Pulses and Inverter Phase Voltages

The PWM gate pulses generated by the controller are shown together with the inverter phase voltages in Figure 4.

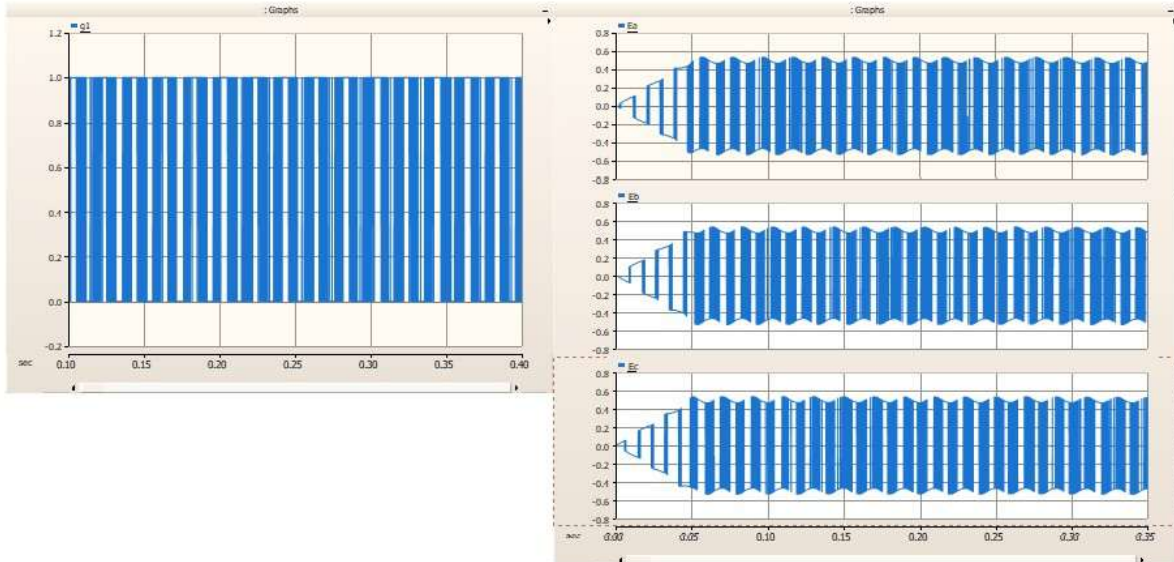


Figure 4: PWM gate pulses and three-phase inverter output voltages.

The gate pulses control the switching of the three-phase inverter. Because of the switching operation, the inverter phase voltages contain high-frequency PWM components.

13.3 Grid Current, Line Voltage, and d - q Currents

Figure 5 presents the grid current, line voltage, and the controlled d -axis and q -axis current components.

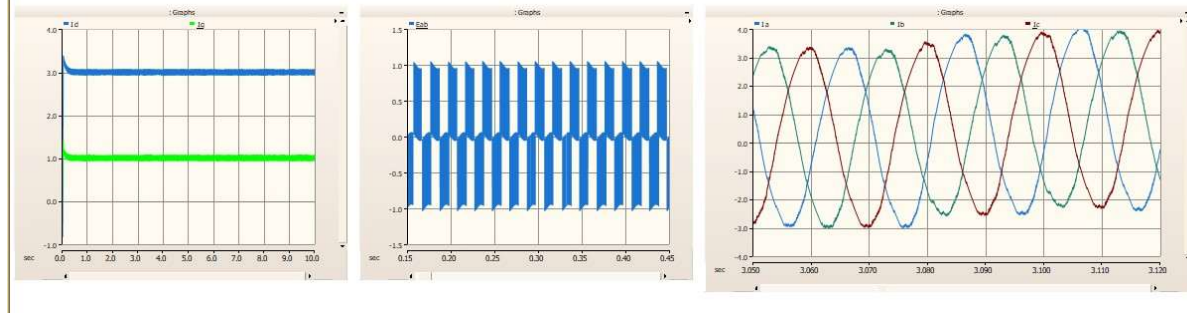


Figure 5: Grid current, line voltage, and d - q current components.

The three-phase currents are balanced and displaced by approximately 120° from each other. The controlled d and q current components represent the active and reactive current control objectives.

13.4 Grid Voltage, Reference Voltage, and Grid Angle

The synchronization-related signals are shown in Figure 6.

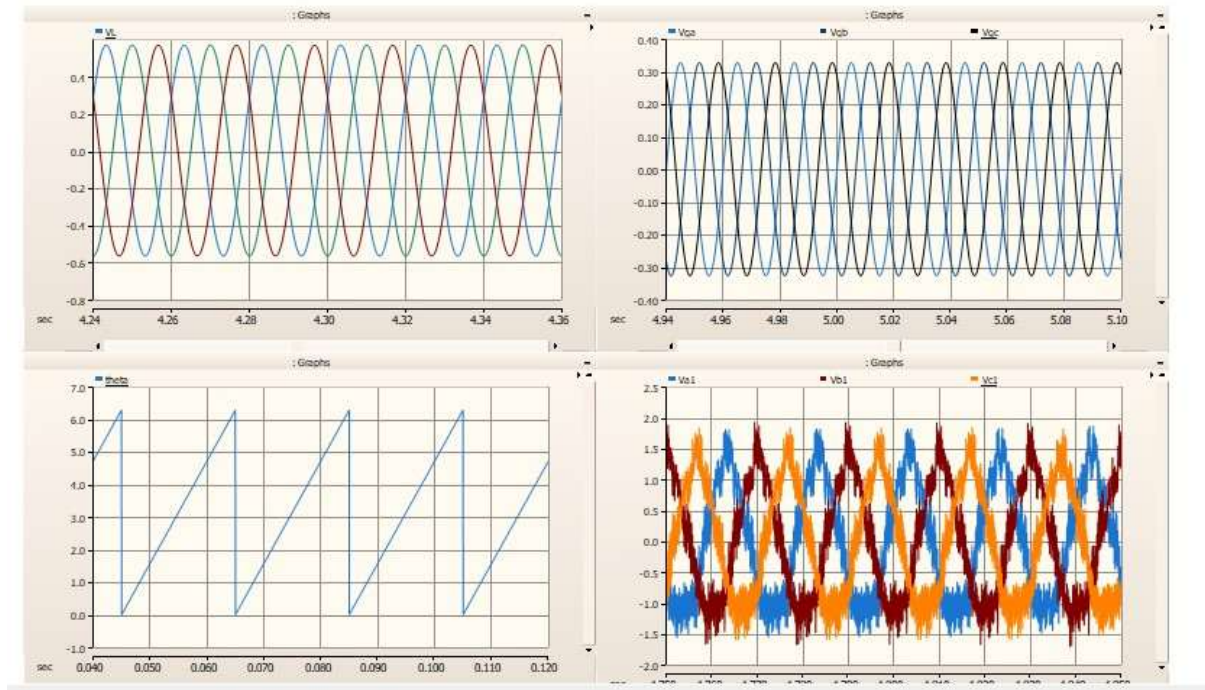


Figure 6: Grid phase voltage, reference voltage, and estimated grid angle.

The estimated angle θ is continuously updated by the PLL and is used by the Park and inverse Park transformations.

13.5 PI Controller Output and Modulating Waves

Figure 7 shows the output of the PI controllers and the generated modulation signals.

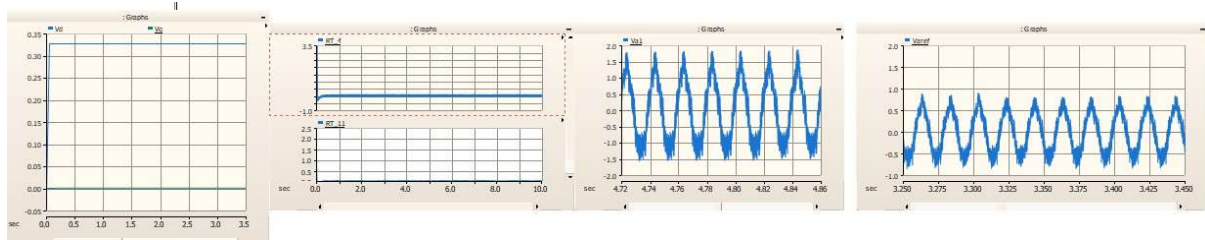


Figure 7: PI controller outputs, v_d , v_q , and modulation signals.

The PI controllers regulate the current errors and generate the reference voltage components required to control the inverter output.

14 Discussion

The simulation demonstrates the complete operation of a three-phase grid-connected inverter with synchronous reference frame control.

The important observations from the simulation are:

- The PLL successfully generates the grid phase angle required for synchronization.
- The three-phase quantities are successfully transformed into the synchronous dq reference frame.
- The d -axis and q -axis current components are independently controlled using PI controllers.
- The controller generates voltage reference signals for the inverter.
- The reference signals are converted into three-phase modulation waves.
- PWM gate pulses are generated for the six inverter switches.
- The inverter produces balanced three-phase output voltages.
- The inverter current is controlled before being injected into the electrical grid.

15 Conclusion

A complete three-phase grid-connected inverter system was modelled and simulated in this project. The system consists of a DC source, a three-phase voltage source inverter, a grid interface filter, and an AC grid.

A synchronous reference frame control strategy was implemented for controlling the grid current. A phase-locked loop was used to synchronize the controller with the grid voltage and generate the grid phase angle.

The three-phase voltage and current quantities were transformed into the dq reference frame. PI controllers were used to regulate the d -axis and q -axis currents. The resulting voltage reference signals were transformed back into three-phase quantities and used to generate PWM switching signals.

The simulation results demonstrate proper generation of inverter phase voltages, PWM gate pulses, balanced grid currents, dq -frame quantities, and synchronization signals.

The developed model provides a fundamental platform for further work on advanced grid-connected converter control, digital implementation on a microcontroller, and the application of advanced converters such as modular multilevel converters.

16 Future Work

The following extensions can be considered:

- Implementation of the controller on a digital signal processor or microcontroller.
- Implementation using TI C2000 controllers.
- Investigation of different PLL techniques.
- Implementation of advanced current controllers.
- Addition of active and reactive power control.
- Investigation of harmonic performance and total harmonic distortion.
- Extension of the control strategy to a modular multilevel converter.
- Hardware implementation of the grid-connected converter.

References

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